

WASP-104b

R-2329

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_200318 · created 2026-08-18T23:08:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458926.8707 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.044
Transit depth (Rp/Rs)^2	1.28 +/- 1.0 [%]
Orbital Inclination (inc)	83.52 +/- 2.87
Airmass coefficient 1 (a1)	1.25 +/- 0.73
Airmass coefficient 2 (a2)	-0.17 +/- 0.21
Scatter in the residuals of the lightcurve fit is	54.37 %
Best Comparison Star	None
Optimal Aperture	6.93
Optimal Annulus	10.04
Transit Duration (day)	0.06 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.113 $\pm$ 0.044 vs 0.1214 (deviation 0.19 σ)
Duration fit vs published	0.06 d vs 0.0742 d (deviation 0.49 σ)
Mid-time O-C	-5.5 min
Depth SNR	2.6
Residual scatter	54.37 %

Light curve

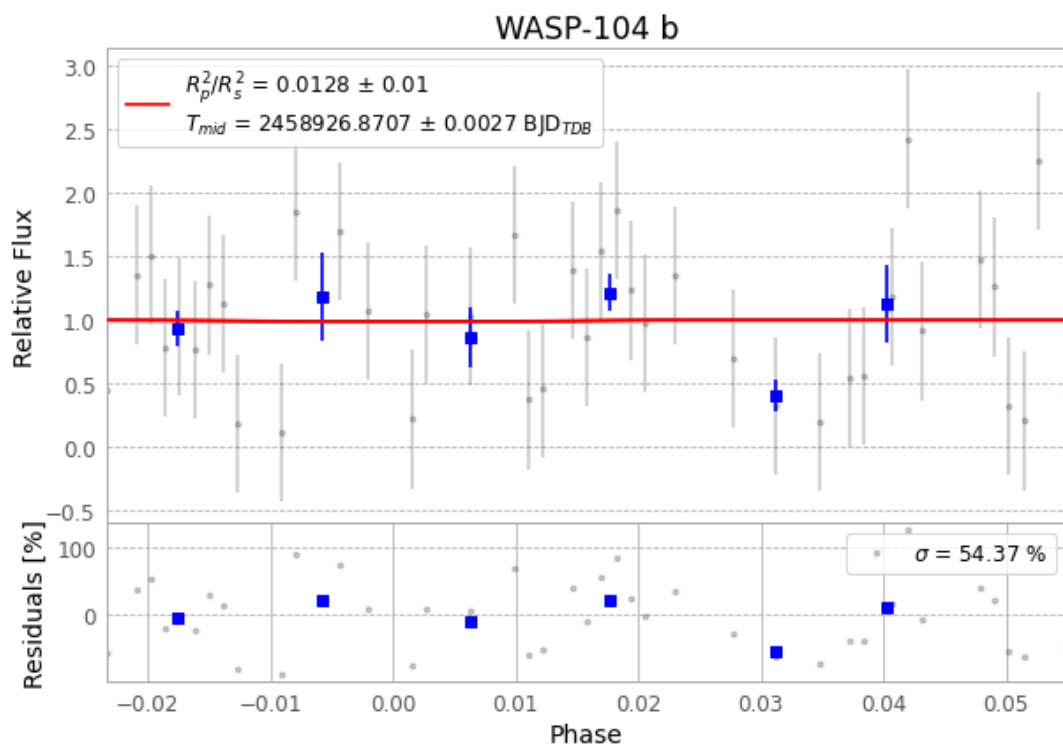


Figure 1 — FinalLightCurve_WASP-104 b_2020-03-18.png (SHA-256 985f1069df6a351f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0341e20786ae0b56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), WASP-104200318074510.FITS → WASP-104200318110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-200318.log (8bf02962ff7d...), FinalLightCurve_WASP-104 b_2020-03-18.png (985f1069df6a...), FinalLightCurve_WASP-104 b_2020-03-18.csv (e2e6fb11a6c8...)

Compute resources

Wall time 164.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:08Z.